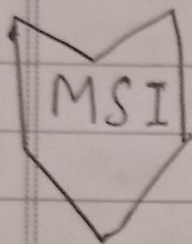




UNIT-1

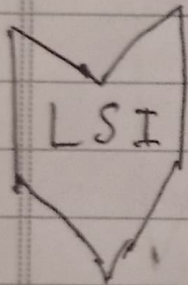
EVOLUTION OF VLSI

Evolution in Logic Complexity in IC's



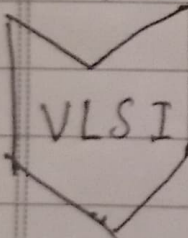
MSI = Medium Scale Integration

- Introduced in 1967
- Logic Block Per chip 20-200



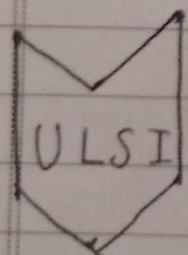
LSI = Large Scale Integration

- Introduced in 1972
- Logic Block Per chip 200-2000



VLSI = Very Large Scale Integration

- Introduced in 1978
- Logic Block Per chip 2000-20000



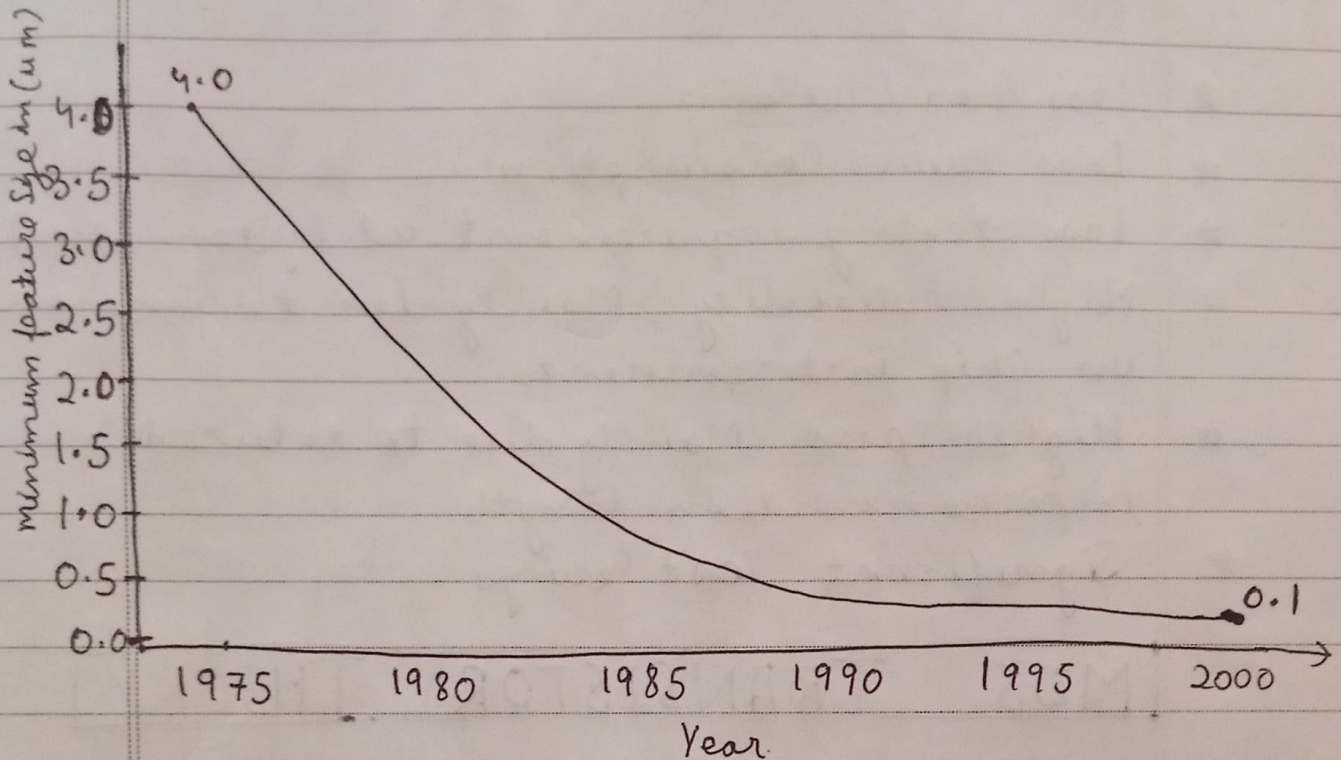
ULSI = Ultra Large Scale Integration

- Introduced in 1989
- Logic Block Per chip 20000 >>

Evolution of Size



Evolution of Size in Integrated Services.



Size of Integrated chip is 4.0 mm in 1975 and that is reduced to 0.1mm in 2000.

Prominent Information Technology Services

Video on Demand

Speech Processing

Wireless/Cellular Data Communication

Data Communication Multimedia Application

Consumer Electronics Portable Computer

Main Frame Computer Personal Computer Computer Network

1975

1980

1985

1990

1995

2000

Remarks

Teacher's Signature:



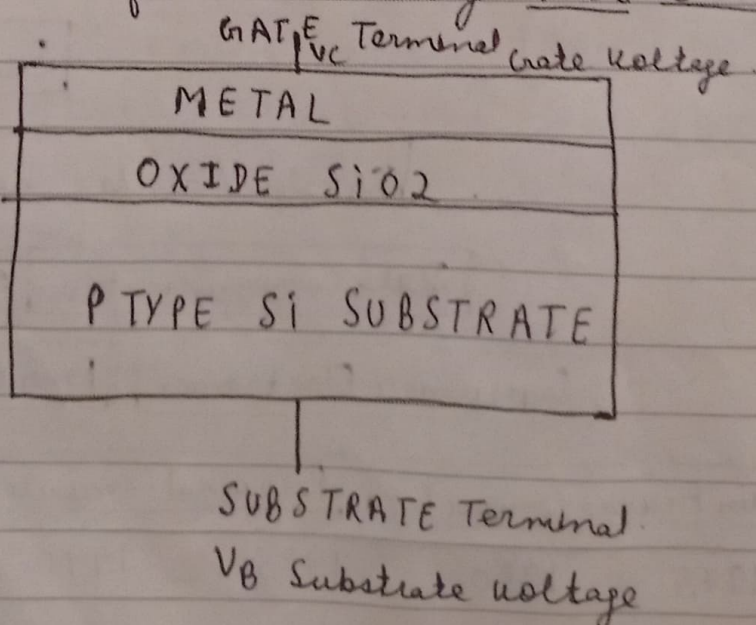
Features of Integrated Ckts

- * Less Area / volume.
- * Less Power Consumption.
- * Less testing requirement at System level.
- * Higher reliability, Mainly due to improved on chip interconnects.
- * Higher Speed Mainly due to reduced interconnection length.
- * Significant Cost Savings.

MOS TRANSISTOR THEORY

Basics of MOS TRANSISTOR

- * It is MOSFET (Metal Oxide Semi-Conductor - Field Effect transistor).
- * It has 4 Terminals : Gate, Substrate, Drain, Source.
- * This device is formed using MOS STRUCTURE



Remarks

Teacher's Signature: _____



- * Structure of MOS transistor is Symmetrical to drain and Source terminal.

Types of MOS Transistor

- * These are 2 types of MOS Transistor based on channel
 - Enhancement Type MOS Transistor
It has no conducting channel region at Zero gate bias voltage.
 - Depletion Type MOS Transistor
It has conducting channel region at zero gate bias voltage.

THRESHOLD VOLTAGE

- * It is a Minimum Voltage V_{GS} (Gate to Source) required to form Inversion layer (channel) in b/w Source & drain
- * Threshold voltage defines operation of MOSFET
- * If V_{GS} is less than threshold voltage then MOSFET will stay in Cut off region.



Parameters of Threshold Voltage.

- * There are 4 Physical parameters which effects the threshold voltage of MOS Structure:
- Work function. d/B gate and the channel.
 - The Gate Voltage Component to change Surface Potential.
 - The Gate Voltage Component to offset the depletion region charge.
 - The Voltage Component to offset the fixed charges in the gate oxide and in the Silicon oxide interface.

Derivation of Threshold Voltage.

- * Work function difference ϕ_{GC} in b/w the gate and channel is given by.

$$\phi_{GS} = \phi_F (\text{Substrate}) - \phi_M (\text{Metal})$$

$$\phi_{GC} = \phi_F (\text{Substrate}) - \phi_F (\text{Gate Polysilicon})$$

- * For Surface Inversion, surface potential should change from ϕ_F to $-\phi_F$. So net change will be $-\phi_F - (\phi_F) = -2\phi_F$

- * So Depletion region charge density at surface inversion ($\phi_s = -\phi_F$) is given by.



$V_{CB} > 0$ (Small) & $V_B = 0$ Depletion region.

- * Here we will calculate depletion width x_d as function of Surface potential ϕ_s
- * Mobile hole charge in a thin horizontal layer parallel to surface is.
$$dQ = -q N_A dx$$
- * The change in Surface potential required to displace thin charge dQ by distance x_d can be calculated by Poisson eqn.

$$d\phi = \frac{-\kappa \cdot dQ}{\epsilon_{Si}}$$

$$d\phi = \frac{q N_A x dx}{\epsilon_{Si}}$$

- * In integration dx varies from 0 to x_d & Potential varies from Fermi Potential ϕ_F to Surface potential ϕ_s

$$\therefore \int_{\phi_F}^{\phi_s} d\phi = \int_0^{x_d} \frac{q N_A x dx}{\epsilon_{Si}}$$

$$\therefore \phi_s - \phi_F = \frac{q N_A x_d^2}{2 \epsilon_{Si}}$$

- * So depth of depletion region is

$$\therefore x_d = \sqrt{\frac{2 \epsilon_{Si} (\phi_s - \phi_F)}{q N_A}}$$



So depletion region charge density is given by

$$\therefore Q = -qN_A x_d$$

$$\therefore \phi = -\sqrt{2qN_A \epsilon_{Si} (\phi_s - \phi_F)}$$

$V_G > 0$ (Large) and $V_B = 0$ Inversion Region

- * when Surface inversion takes place the density of mobile electrons on the surface equals to the density of holes in the P type.
- * Condition for that is $(\phi_s = -\phi_F)$
- * Now if we ~~forth~~ further increase gate voltage V_G then width of depletion region will not increase but it increase the no. of electron
- * So Maximum depletion region depth can be calculated by

$$\therefore x_{dm} = \sqrt{\frac{2\epsilon_{Si} (2\phi_F)}{qN_A}}$$

- * So Depletion ~~gate~~ region charge density at Surface inversion $(\phi_s = -\phi_F)$ is given by

$$Q_{B0} = -\sqrt{2qN_A \epsilon_{Si} | -2\phi_F |}$$

- * If Substrate bias is given by width V_{SB} the depletion region charge density is given by



$$Q_B = -\sqrt{2qNA\epsilon_{Si}} | -2\phi_F + V_{SB} |$$

- ★ So Component of Voltage which Nullify Value will be $-Q_B/C_{ox}$ where C_{ox} is gate oxide Capacitance per unit area.
- ★ There is also fixed positive charge density Q_{ox} it is due to lattice imperfections at interface
- ★ So Farth Component for threshold Voltage ~~to get~~ will be $-Q_{ox}/C_{ox}$. . .
- ★ So For Zero Substrate bias threshold Voltage is given by

$$V_{T0} = \phi_{GC} - 2\phi_F - \frac{Q_{B0}}{C_{ox}} - \frac{Q_{ox}}{C_{ox}}$$
- ★ For Non-Zero Substrate bias threshold Voltage is

$$V_T = \phi_{GC} - 2\phi_F - \frac{Q_B}{C_{ox}} - \frac{Q_{ox}}{C_{ox}}$$

$$V_T = \phi_{GC} - 2\phi_F - \frac{Q_{B0}}{C_{ox}} - \frac{Q_{ox}}{C_{ox}} - \frac{Q_B - Q_{B0}}{C_{ox}}$$

$$V_T = V_{T0} - \frac{Q_B - Q_{B0}}{C_{ox}}$$

★ Here

$$\frac{Q_B - Q_{B0}}{C_{ox}} = \frac{-\sqrt{2qNA\epsilon_{Si}}}{C_{ox}} (\sqrt{|-2\phi_F + V_{SB}|} - \sqrt{|-2\phi_F|})$$

- ★ So Threshold Voltage is

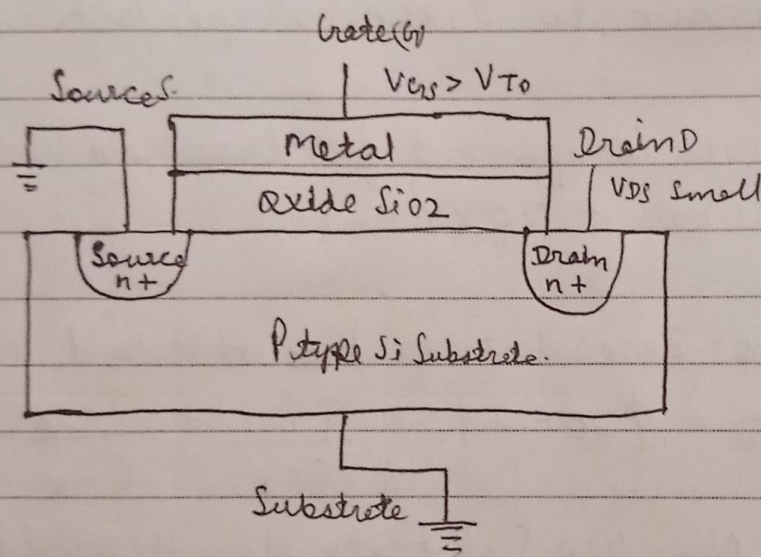
$$V_T = V_{T0} + V (\sqrt{|-2\phi_F + V_{SB}|} - \sqrt{|-2\phi_F|})$$



where, V is substrate bias or body effect coefficient

$$V = \sqrt{\frac{2qNAE_{Si}}{C_{ox}}}$$

Gradual channel Approximation in MOSFET



- * $V_{GS} > V_{TO}$ and $V_{GD} > V_{TO}$ to form channel.
- * Threshold Voltage V_{TO} is assumed to be constant
- * Channel Voltage $V_c(y)$ will change with respect to y
 - $V_c(y=0) = V_S = 0$
 - $V_c(y=L) = V_{DS}$
- * Electric field Component in y direction is dominant compared to x direction, so current flow only confined in y direction.
- * Let $Q_i(y)$ is charge density in channel.



$$Q_1(Y) = -C_{ox} [V_{GS} - V_c(Y) - V_{T0}]$$

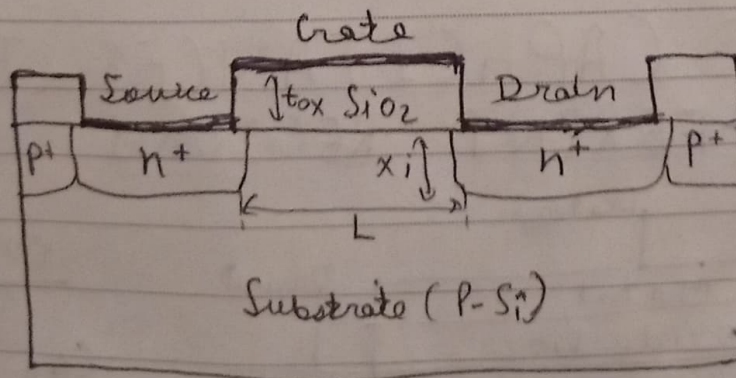
MOSFET CAPACITANCES

Basics

- * Speed of IC is limited by capacitances.
- * These capacitances are not lumped but distributed.
- * It's values can be calculated by 3 dimensional overview of MOSFET.
- * Here, we have already studied C_{ox} which is gate oxide capacitance its unit is F/cm^2 .

$$C_{ox} = \frac{\epsilon_{ox}}{t_{ox}}$$

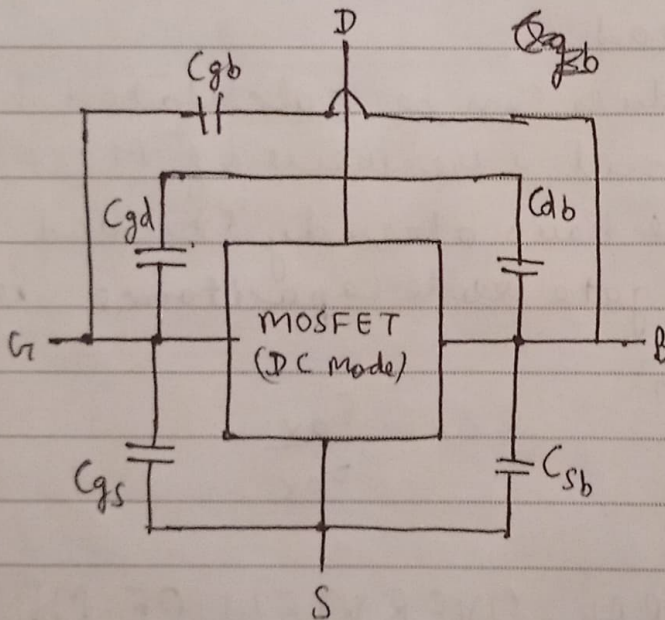
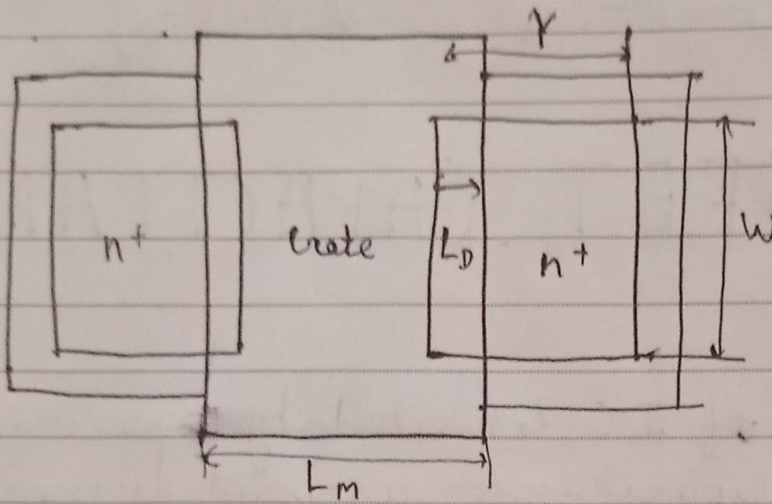
STRUCTURAL OVERVIEW OF MOSFET CAPACITANCES



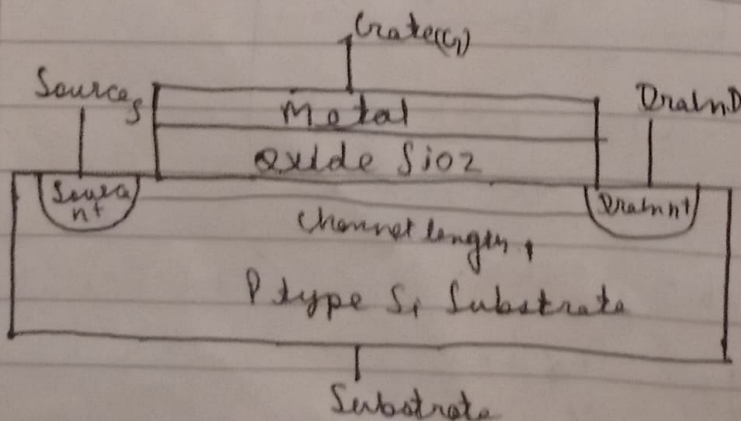
$$C_{gb} = C_{ox} L W$$

$$C_{gd} = C_{ox} L W$$

$$C_{gs} = C_{ox} L W$$



MOSFET CAPACITANCES IN DIFFERENT REGIONS



CUTT OFF Region

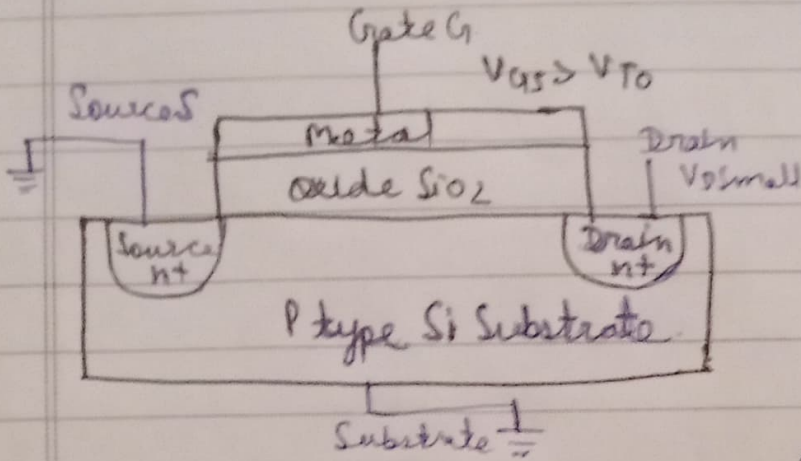
$$C_{GB} = C_{ox} W L$$

$$C_{GD} = C_{ox} W L_D$$

$$C_{GS} = C_{ox} W L_S$$

Remarks

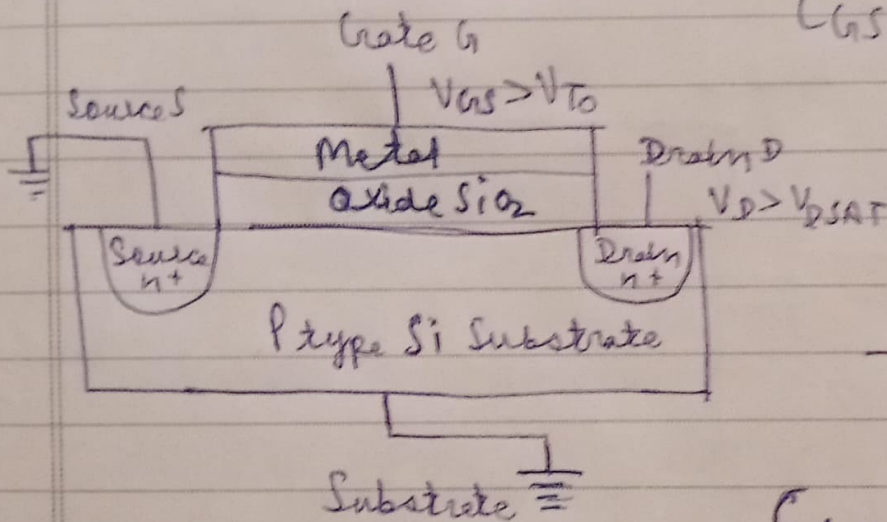
Teacher's Signature:

LINEAR Region

$$C_{GB} = 0$$

$$C_{GD} = \frac{1}{2} C_{OX} W_L + C_{OX} W_{LD}$$

$$C_{GS} = \frac{1}{2} C_{OX} W_L + C_{OX} W_{LD}$$

SATURATION Region

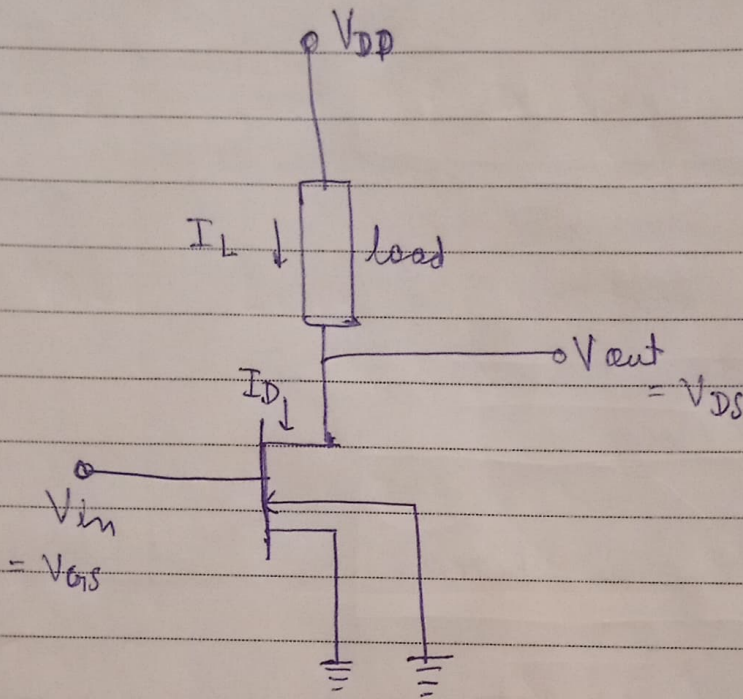
$$C_{GB} = 0$$

$$C_{GD} = C_{OX} W_{LD}$$

$$C_{GS} = \frac{2}{3} C_{OX} W_L + C_{OX} W_{LD}$$



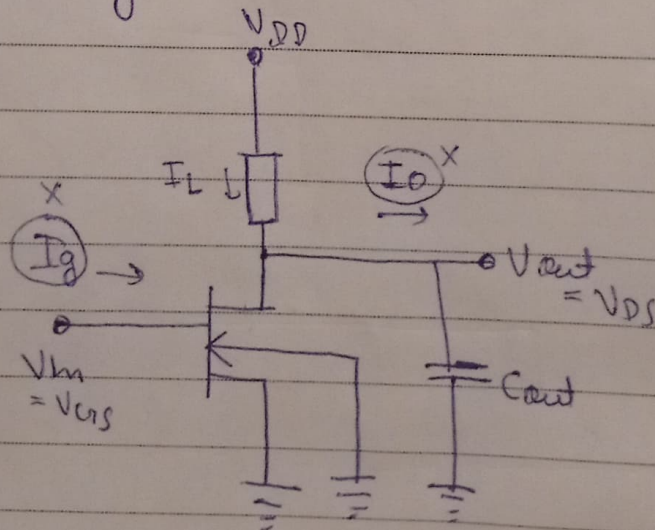
NMOS INVERTER



Circuit Diagram

Working

⇒ Due to capacitance at gate, No gate current is flowing.



Remarks

Teacher's Signature: _____



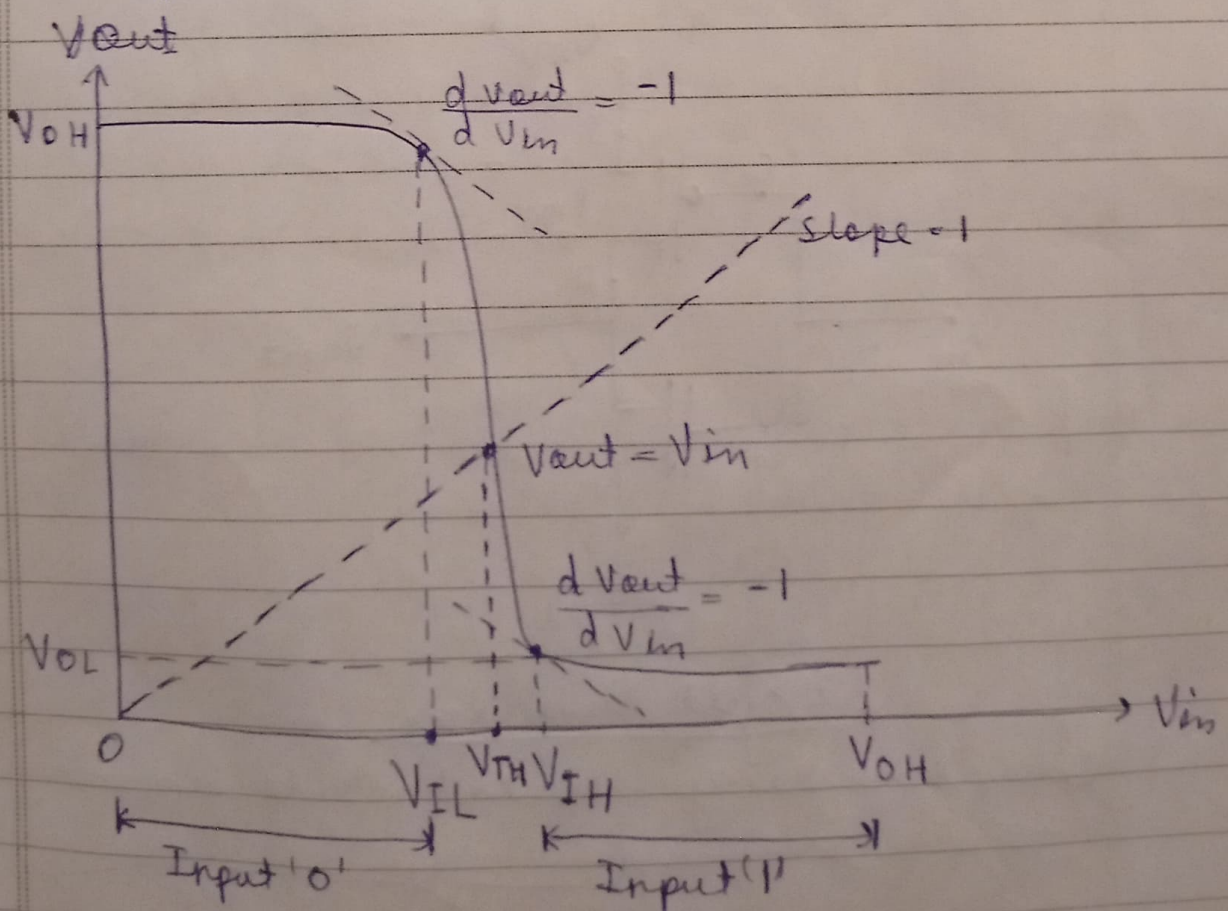
- Hence $I_L = I_D$
- If V_{in} is less, No inversion inversion layer is there, I_D is minimum,
 $V_{out} = V_{DD} - I_L \times \text{load}$
 $O/P \approx V_{DD}$

- If we further increase V_{in} , it will form inversion layer. So I_D current will increase so

$$V_{out} = V_{DD} - I_L \times \text{load}$$

O/P will decrease.

Transfer Voltage Characteristics





V_{OH} = High O/P voltage.

= It is Max O/P, when O/P is logic 1.

V_{OL} = Low O/P voltage.

= It is Min O/P when O/P is logic 0.

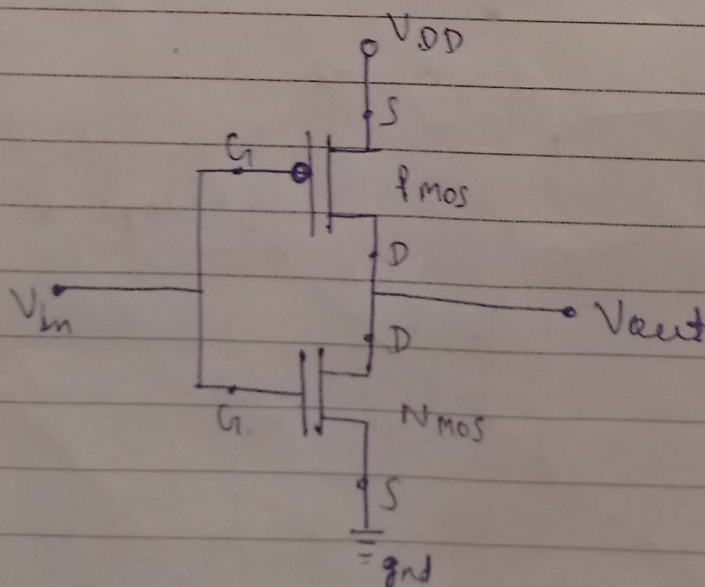
V_{IL} = Low I/P voltage.

= It is Max I/P when I/P is logic 0.

V_{IH} = High I/P voltage;

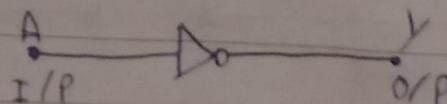
= It is min I/P when I/P is logic 1.

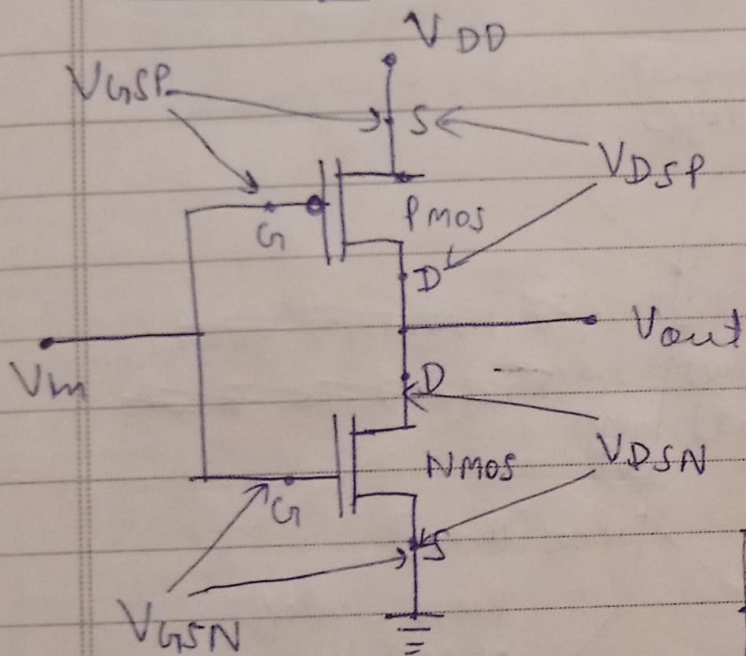
CMOS INVERTER



Circuit Diagram

Complementary Metal Oxide Semiconductor.



WorkingFor PMOS

If I/P 0, S to D = S-C

If I/P 1, S to D = O-C

For NMOS

If I/P 0, D to S = O-C

If I/P 1, D to S = S-C

I/P	PMOS	NMOS	O/P
0	S-C	O-C	V_{DD}
1	O-C	S-C	gnd(0)

$$\star V_{TN} (+ve)$$

$$\star V_{TP} (-ve)$$

$$\star V_{GSN} = (V_G - V_S)_N = V_{in}$$

$$\star V_{DSN} = (V_D - V_S)_N = V_{out}$$

$$\star V_{GSP} = (V_G - V_S)_P = V_{in} - V_{DD}$$

$$\star V_{DSP} = (V_D - V_S)_P = V_{out} - V_{DD}$$

In NMOS

$$\rightarrow V_{GSN} < V_m \text{ [cutoff]}$$

$$\rightarrow V_{GSN} > V_m \text{ [linear/sat]}$$

$$V_{GSN} - V_{TN} > V_{DSN} \text{ [linear]}$$

$$V_{GSN} - V_{TN} < V_{DSN} \text{ [sat]}$$

In PMOS

$$- V_{GSP} > V_{TP} \text{ [cutoff]}$$

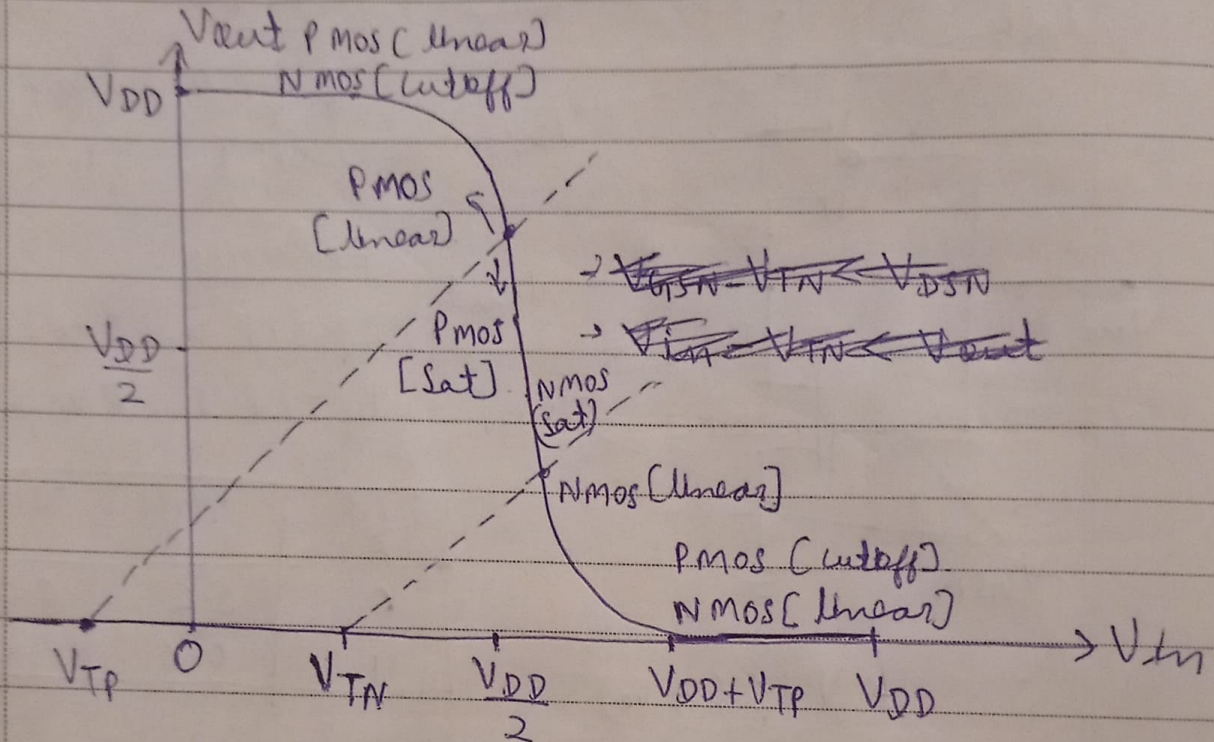
$$- V_{GSP} < V_{TP} \text{ [linear/sat]}$$

$$V_{GSP} - V_{TP} > V_{DSP} \text{ [sat]}$$

$$V_{GSP} - V_{TP} < V_{DSP} \text{ [linear]}$$



Voltage Transfer characteristics



$$\Rightarrow V_{GSN} - V_{TN} < V_{DSN}$$

$$\Rightarrow V_{in} - V_{TN} < V_{out}$$

$$\Rightarrow V_{GSP} - V_{TP} > V_{DSP}$$

$$\Rightarrow V_{in} - V_{DD} - V_{TP} > V_{out} - V_{DD}$$

$$\Rightarrow V_{in} - V_{TP} > V_{out}$$

PULL UP & PULL DOWN RATIO

Pull up = A Network that provides a low resistance path to V_{DD} when O/P is logic '1' and provides a high resistance path to V_{DD} when O/P is logic '0' and otherwise.



Pull Down = A Network that provides a low resistance path to GND when output is logic '0' and provides a high resistance to GND otherwise.

POWER DISSIPATION

Power Dissipation can be defined as the product of total current supplied to the circuit and total voltage less or leakage current. when it comes to portability of devices, power dissipation is an unavoidable constraint.

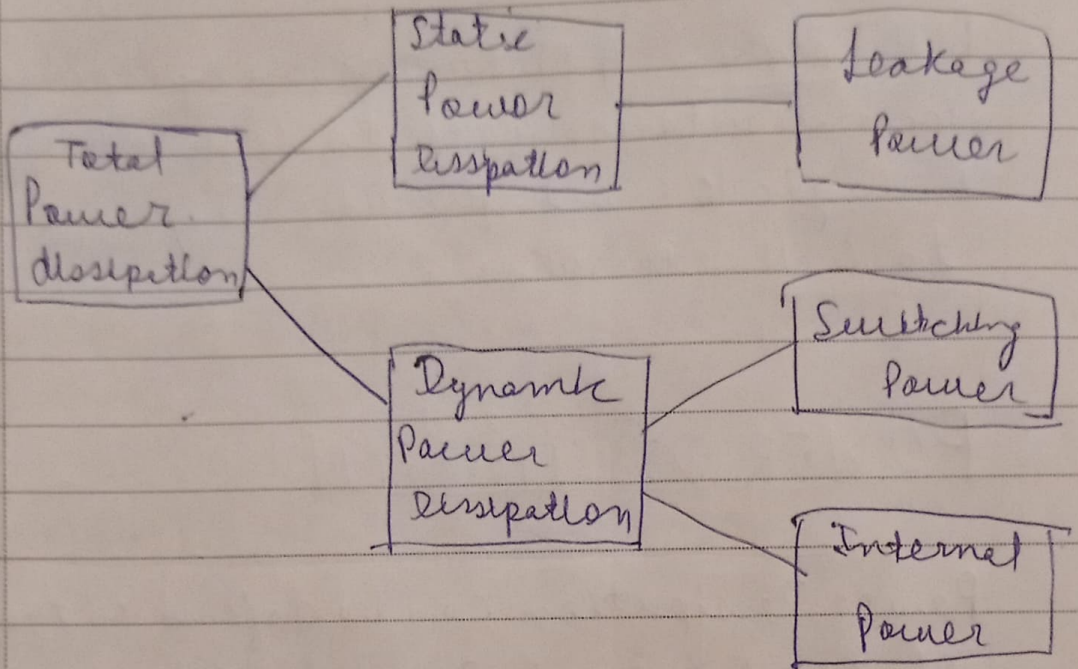
Two types of Power dissipation:-

1) Static Power Dissipation

The power dissipation occurs in form of leakage current when the system is not powered or is in standby mode. ~~In this~~ there are two (i.e) when it is static

2) Dynamic Power Dissipation

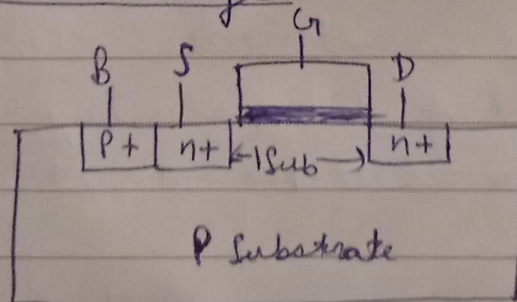
Logic transitions cause logic gates to charge and discharge load capacitance. In other words, this type of power dissipation occurs due to switching activities of transistor.



Various type of leakage in Static Power

- Subthreshold leakage.
- Gate leakage
- Junction leakage.

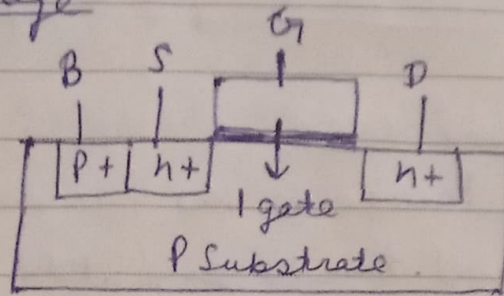
Subthreshold leakage



Sub-threshold leakage current is the current which flows b/w Drain and Source when $V_{gs} < V_t$ (For NMOS) or $V_{sg} < |V_t|$ (PMOS)

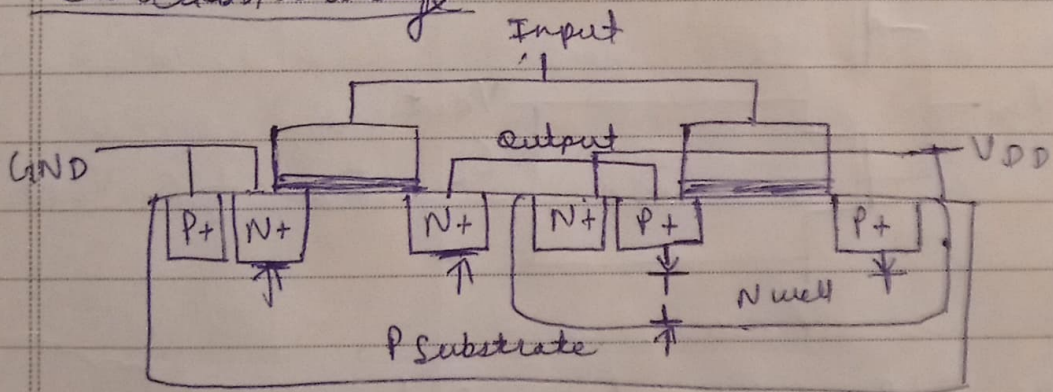


Gate leakage



Leakage through gate is observed when gate oxide is very thin and it allows current to pass through it..

Junction leakage



P-N Junction In a transistor forms diodes. These diodes are reverse biased. However, reverse biased diodes still conduct a small amount of current, which causes Junction leakage current.

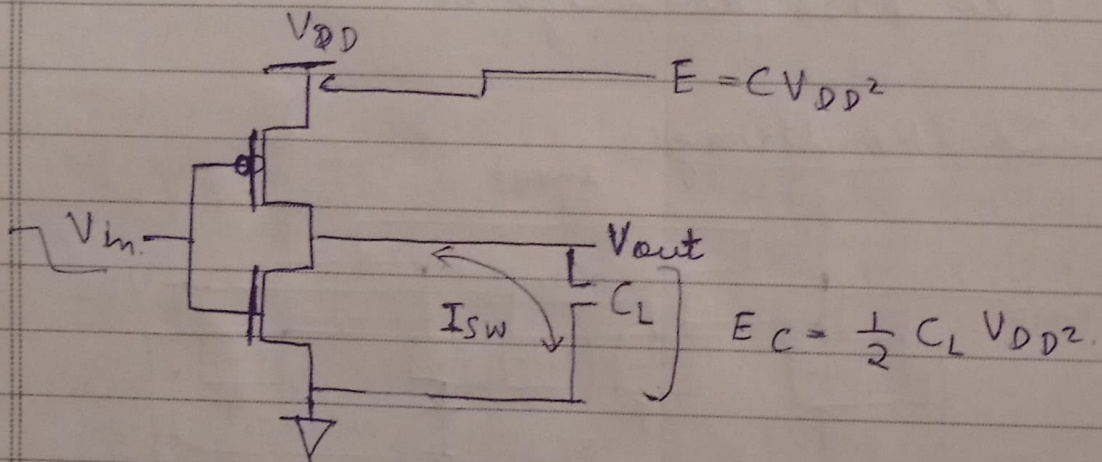


Dynamic Power are of four types.

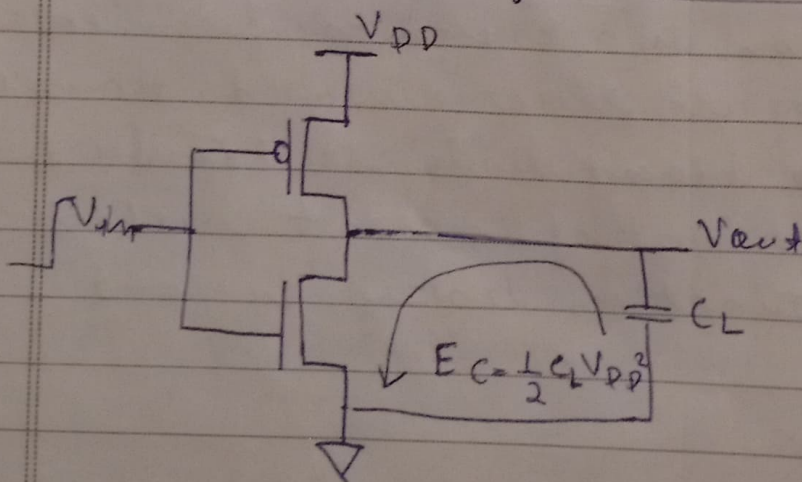
★ Switching Power

Switching Power is the power dissipated while charging and discharging the load capacitor at the output of the cell.

when I/P Switches from 1 to 0



when I/P Switches from 0 back to 1





* Internal Power

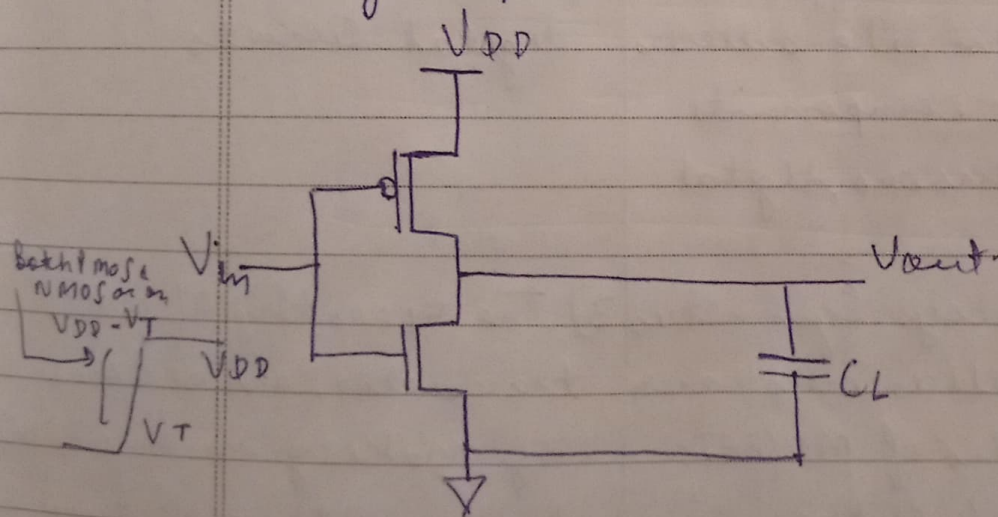
Internal power is any power which is dissipated within the boundary of a cell.

It is also divided into 2 parts

- 1) Short Circuit Power
- 2) Power Needed to charge or ~~etc~~ discharge internal nodes of transistor.

Short Circuit Power

Short ckt power is power dissipated when both PMOS and NMOS are on simultaneously during input transition.



PMOS is ON when: $V_{SS} \leq V_i \leq V_{DD} - V_T$

NMOS is ON when: $V_T \leq V_i \leq V_{DD}$

$\therefore$ both PMOS and NMOS are on for
 $V_T \leq V_i \leq V_{DD} - V_T$



CMOS & NMOS Process Technology

Comparison b/w CMOS & NMOS technology

CMOS

- 1) CMOS stands for Complementary Metal oxide Semiconductor.
- 2) This technology is used to make ICs which are used in different applications like batteries, electronic components, image sensors, digital cameras.
- 3) CMOS employs Symmetric - Cal as well as Complementary pairs of MOSFETs like P-type & N-type MOSFETs for the operation of logic functions.

- 4) The modes of operation of CMOS are accumulation like depletion & inversion.

NMOS

- 1) NMOS stands for N-type Metal oxide Semiconductor.

- 2) NMOS technology is used to make logic gates as well as digital circuits.

- 3) The operating of NMOS transistor can be done by making an inversion layer within a P-type transistor body.

- 4) NMOS has 4 modes of operations that simulate other types of MOSFETs like a cut off, triode,



Saturation and Velocity Saturation

- | | |
|--|--|
| <p>5) The CMOS characteristics are low static power consumption as well as high noise immunity.</p> <p>6) CMOS is used in digital logic ckts like microprocessors, SRAM & Microcontrollers.</p> <p>7) The CMOS logic level is 0/5V.</p> <p>8) The Transmission time of CMOS is $t_r = t_f$.</p> <p>9) Layout of CMOS is more regular.</p> <p>10) Load or drive ratio is 1:1 / 2:1.</p> <p>11) Packing density is less. 2N device for N- I/Ps.</p> <p>12) Power Supply May change from 1.5 to 15V.</p> | <p>5) The NMOS characteristics are when voltage increases on top electrode, then electrons attraction will be toward surface.</p> <p>6) NMOS is used to implement digital ckts as well as logic gates.</p> <p>7) The NMOS logic level mainly depends on beta ratio as well as poor noise margins.</p> <p>8) Transmission time is $t_r > t_f$.</p> <p>9) Layout of NMOS is irregular.</p> <p>10) Load or Drive ratio is 4:1.</p> <p>11) Packing density is denser. N+1 device for N- I/Ps.</p> <p>12) Power Supply is fixed based on VDD.</p> |
|--|--|

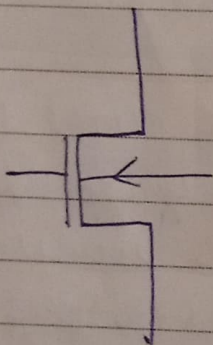


- 13) Transmission gate of CMOS will pass both logic well.
- 14) Power dissipation is zero in standby

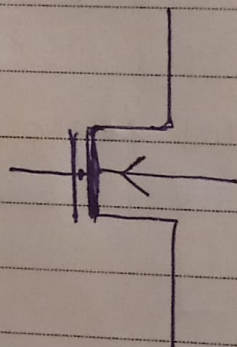
13) only Pass '0' well
Pass '1' will have V_T drop.

14) In NMOS, when output is '0' then power dissipates.

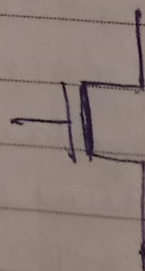
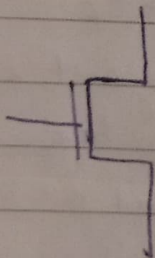
NMOS Fabrication



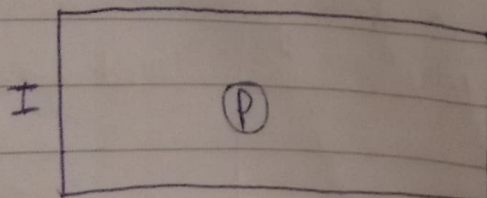
NMOS Enhancement



NMOS Depletion



①



Substrate

Wafer = 75 to 150 mm

0.4 mm Thick

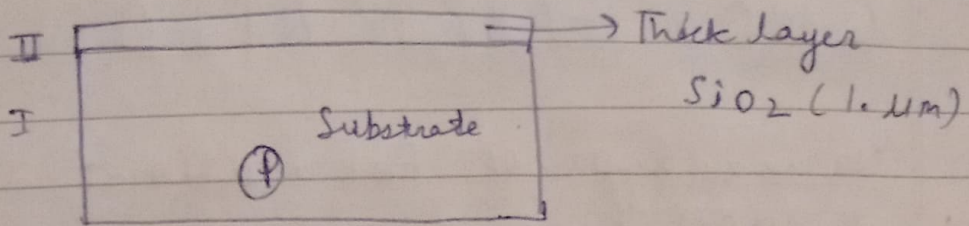
Impurity $\leq 10^{15}/\text{cm}^3$ to $10^{16}/\text{cm}^3$

Remarks

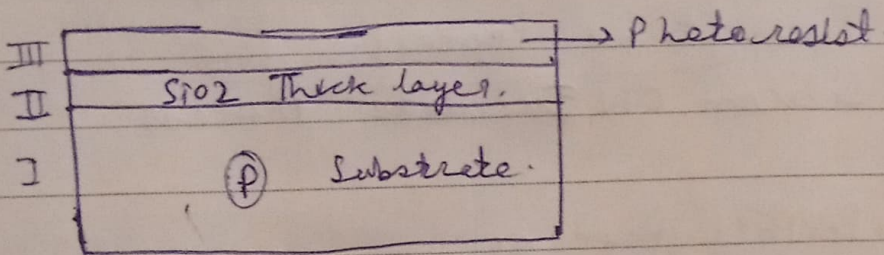
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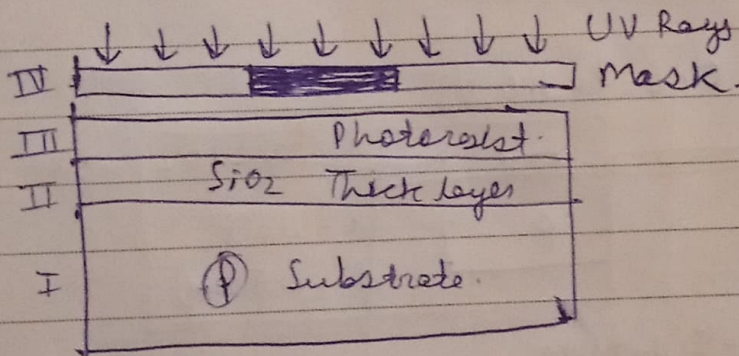
(2)



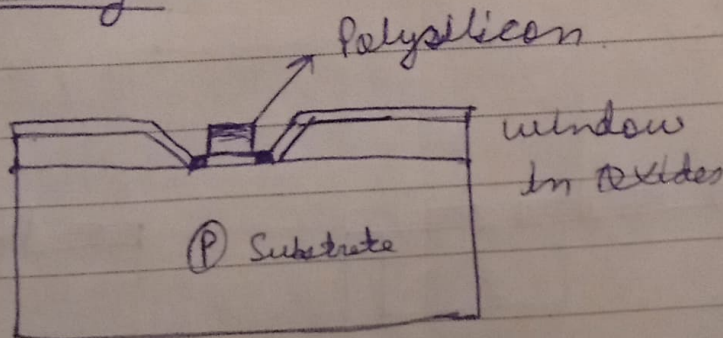
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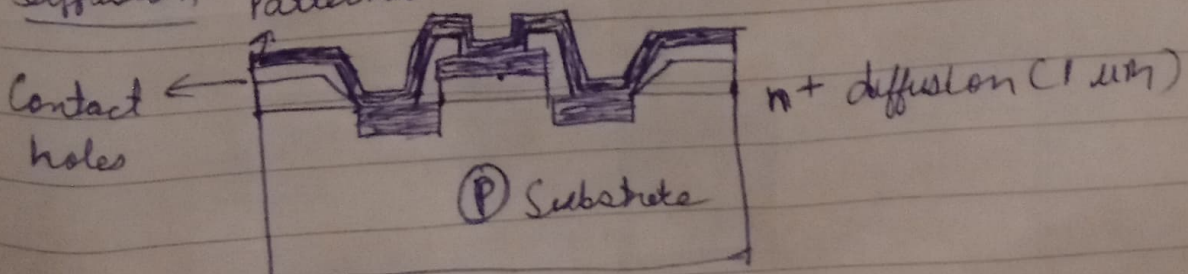
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Etching



Diffusion Patterned Metallization





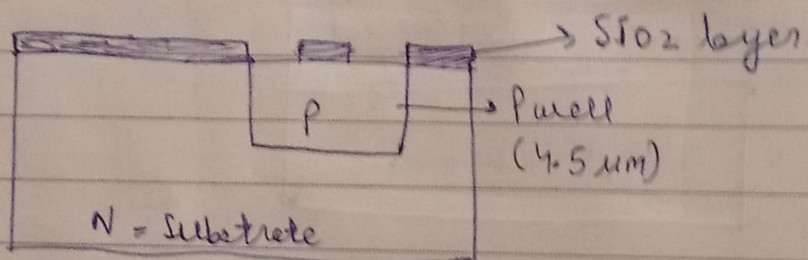
C.MOS FABRICATION

Various types of Fabrication:

- 1) P-Well Process
- 2) N-well Process
- 3) Twin-tube Process
- 4) Silicon ON Insulator Process

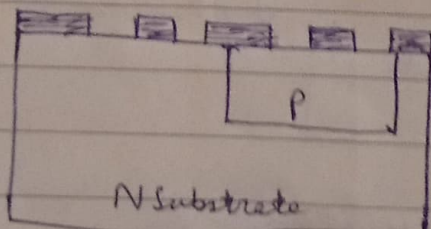
* P-WELL Process

$$V_t = 0.6 \text{ to } 1.0 \text{ V}$$

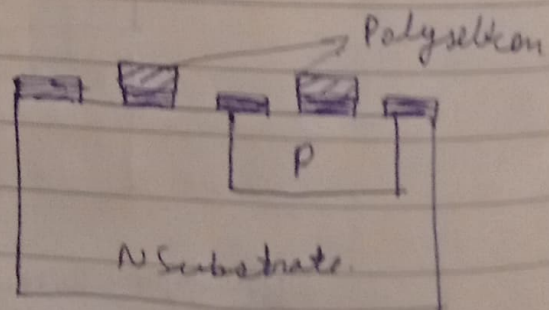


Applying SiO_2 layer

Etching

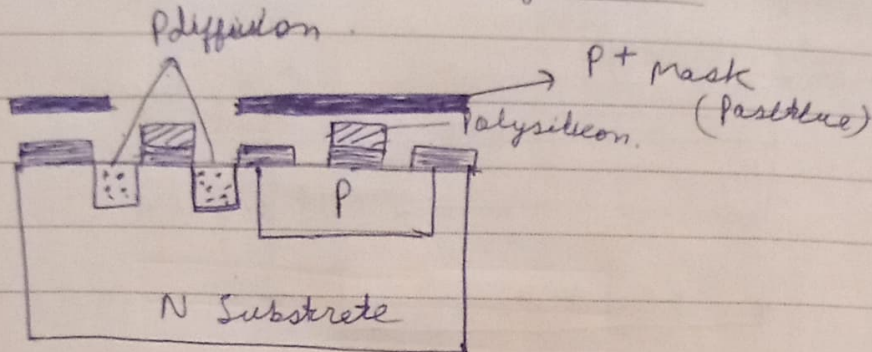


Apply polysilicon

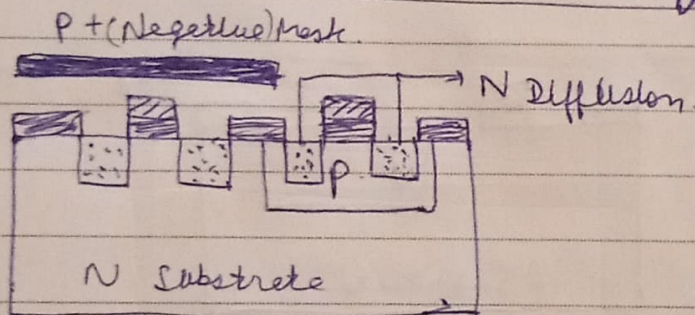




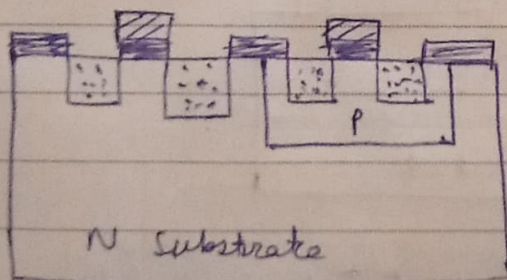
Apply P⁺ mask & P⁺ Diffusion



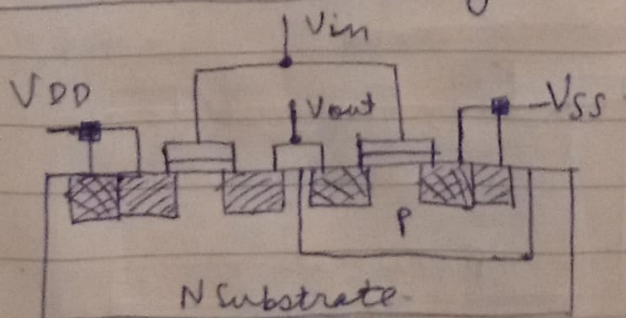
Remove P⁺ mask. and Apply P⁺ (Negative) mask and N Diffusion



Remove P⁺ (Negative) mask



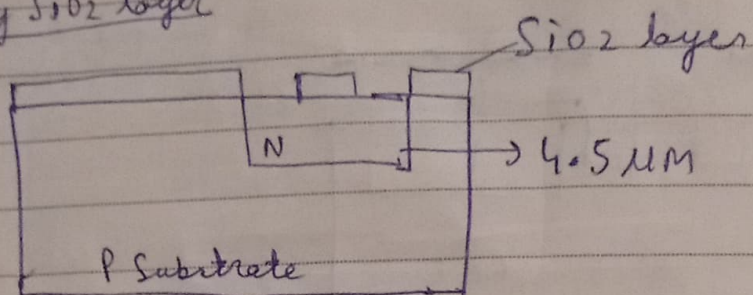
CMOS Power inverter showing V_{DD} & V_{SS} Substrate



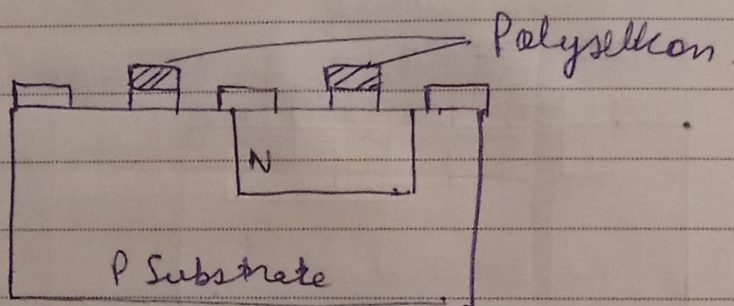


* N-WELL Process

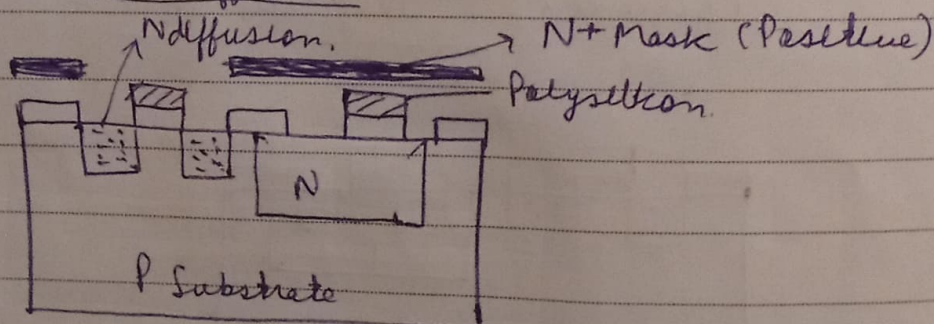
Apply SiO_2 layer



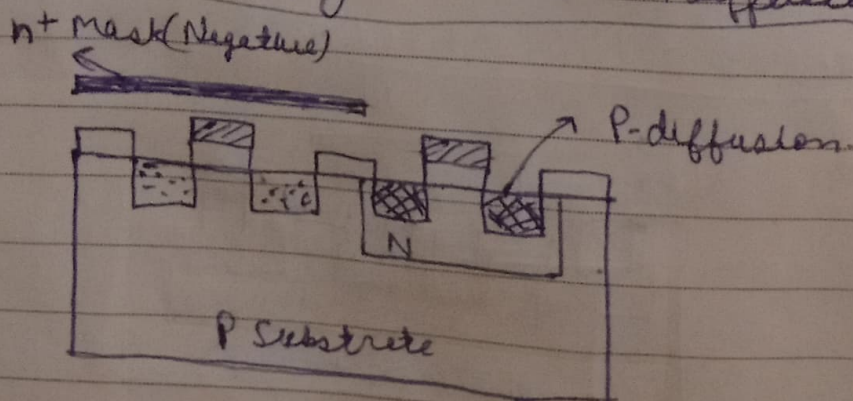
Apply Polysilicon



Apply n^+ Mask & N diffusion



Remove n^+ mask & apply n^+ mask (Negative) & P diffusion

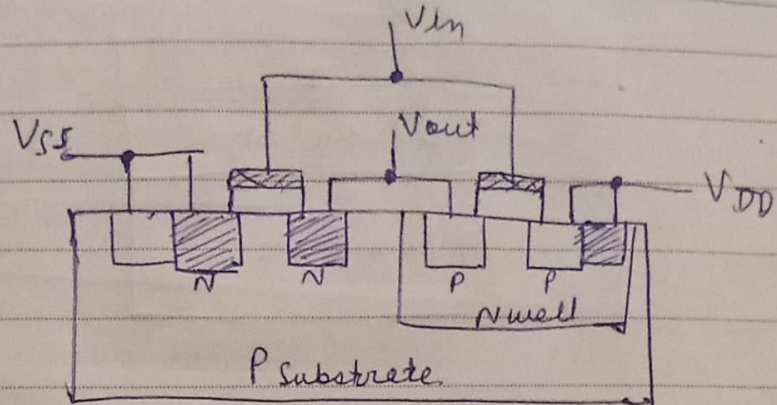


Remarks

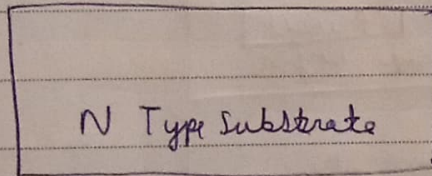
Teacher's Signature:



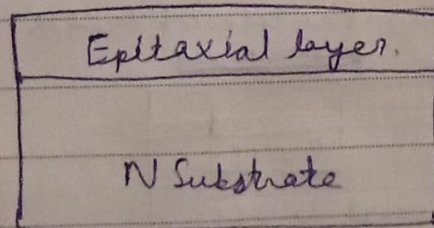
Cross Section View of N-well CMOS Inverter



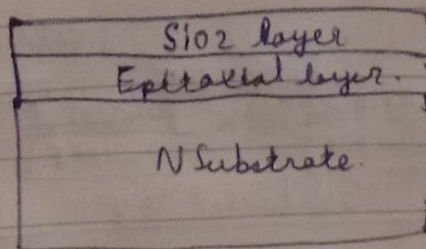
★ TWIN TUBE Process



Apply Epitaxial layer

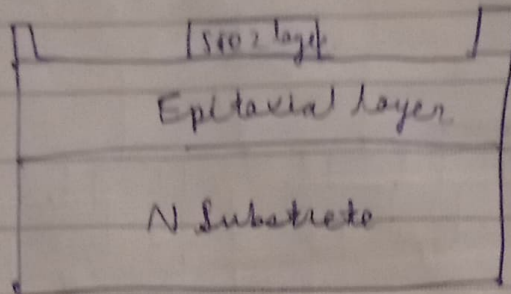


Apply SiO2 layer

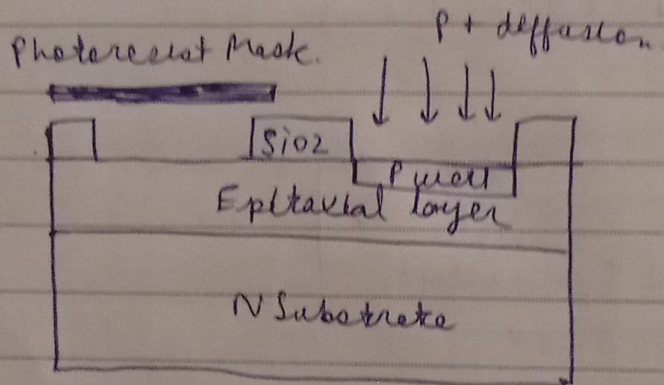




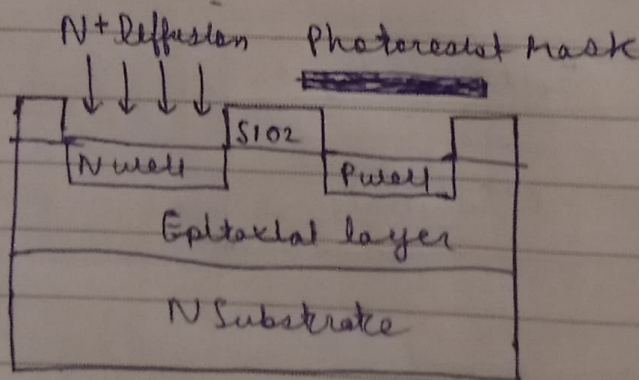
Etching Etching



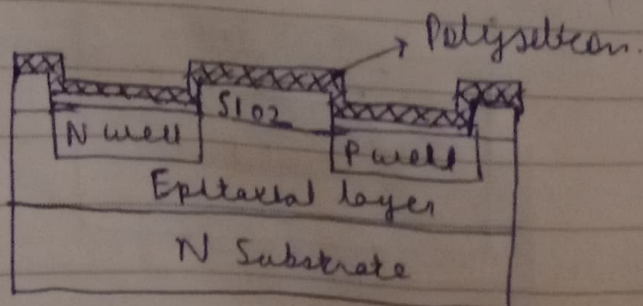
Apply Photoresist mask & P⁺ diffusion



Apply N⁺ diffusion

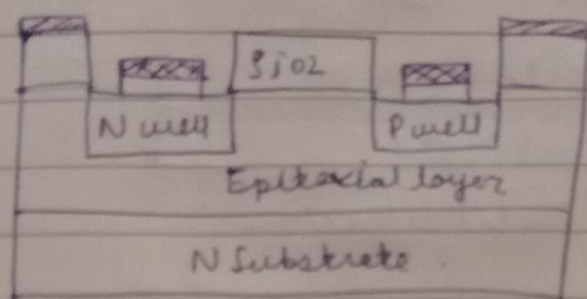


Apply Polysilicon

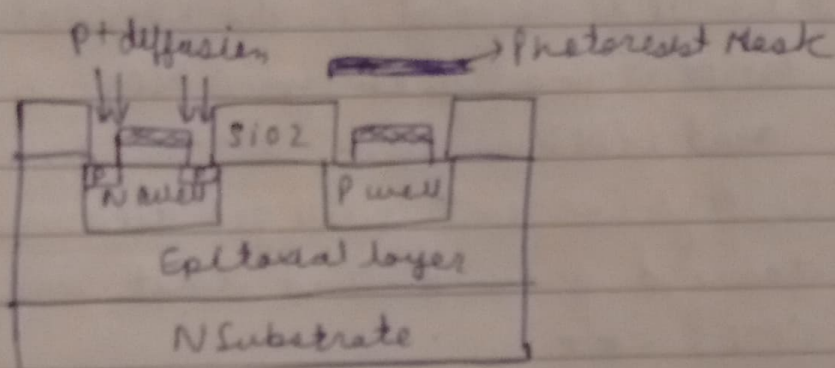




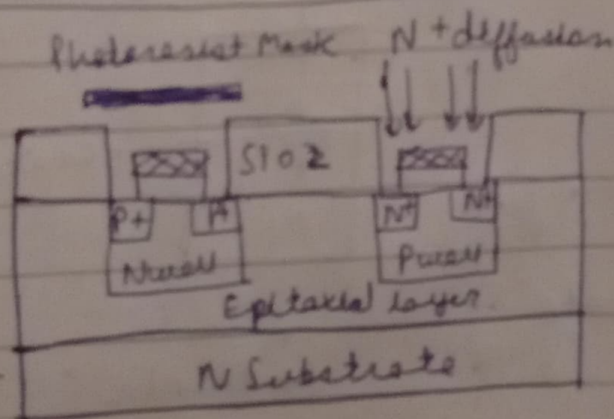
Etching

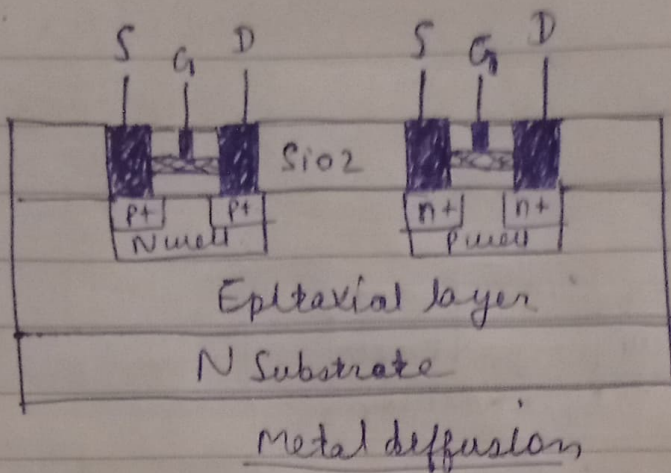


Apply Photoresist Mask & P⁺ Diffusion



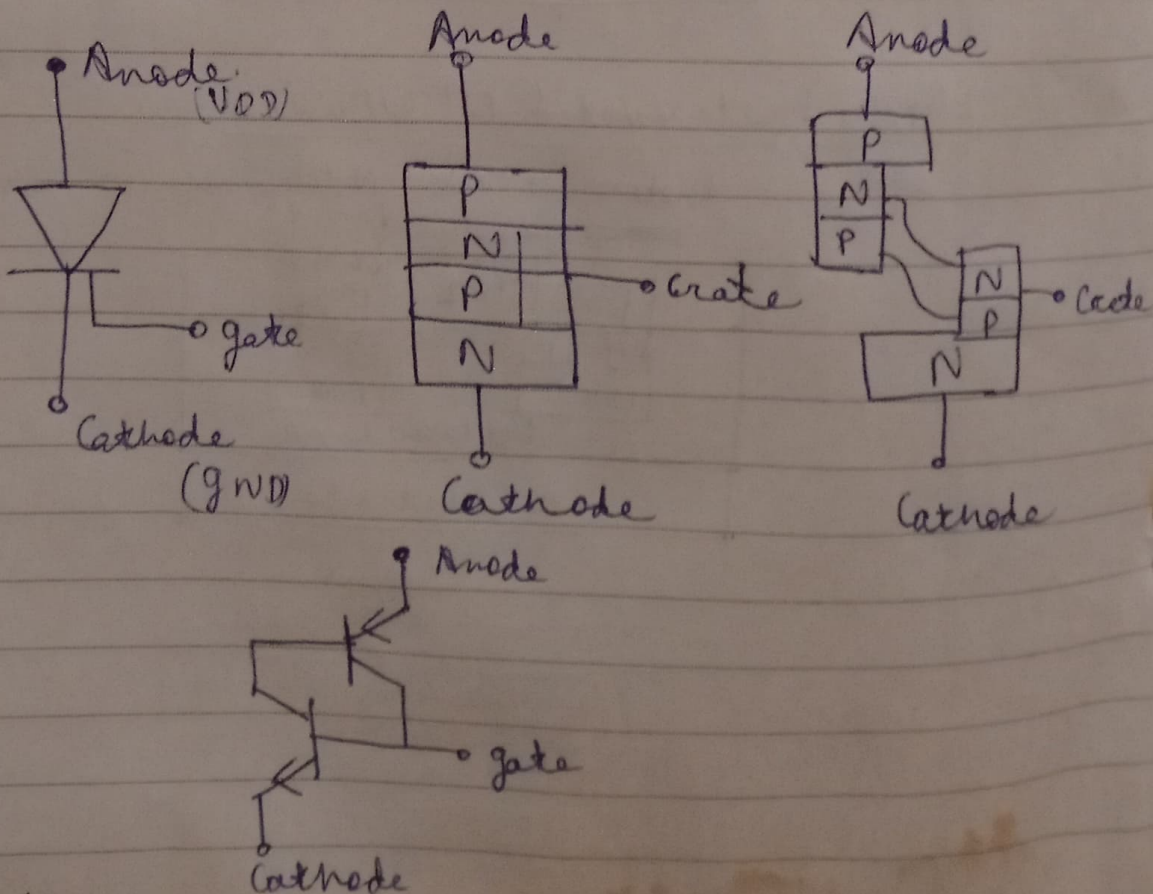
Apply Photoresist & N⁺ Diffusion





LATCH UP In CMOS

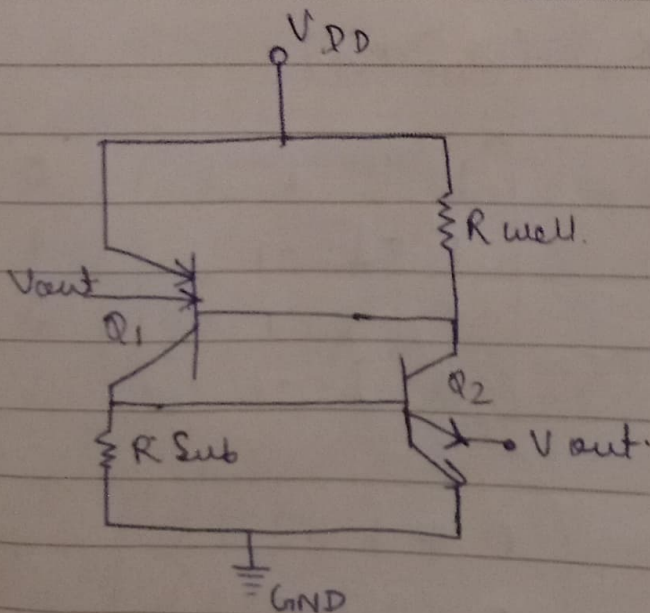
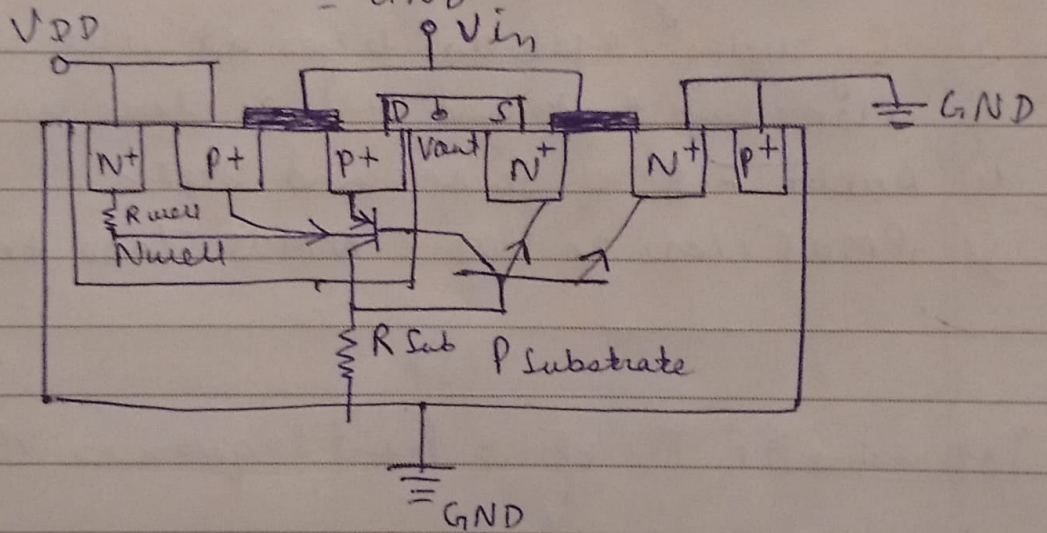
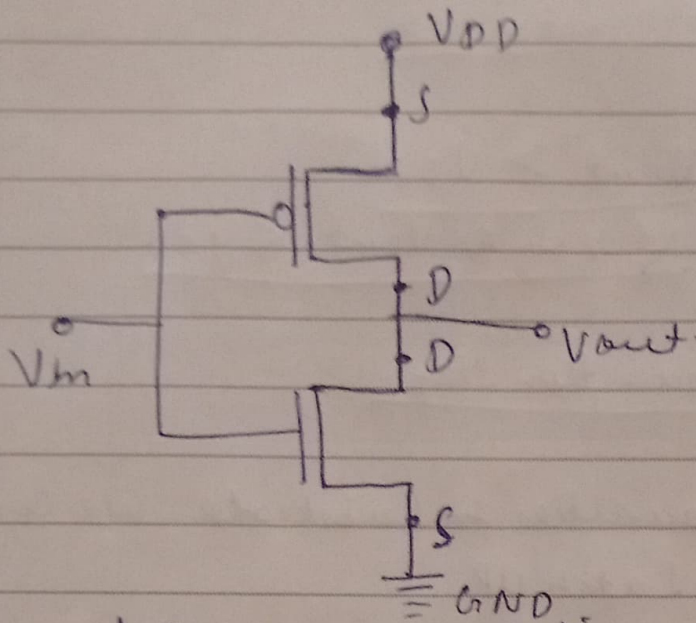
- ⇒ Latch up is defined as the generation of low impedance path in CMOS b/w V_{DD} (Supply) and GND.
- ⇒ So, High Current will get driven by V_{DD} (Supply) toward ground and that may damage CMOS ckt.





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Date : / /





Conditions for Latch UP

Condition 1

If V_{in} or $V_{out} > V_{DD}$

Condition 2 $\equiv$ If V_{in} or $V_{out} < \text{gnd}$

Latch-UP prevention steps

- 1) By gold impurities in substrate. we can minimize Latch UP.
- 2) By using epitaxial layer at substrate.
- 3) Place substrate and well as close as possible.
- 4) Avoid F.B of source and drain.
- 5) PMOS close to V_{DD} & NMOS close to GND.